

Shared Label, Shared Reading, Shared Specification: A Mathematical-Basis Note

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Abstract

This note isolates a reading-point example removed from Draft v3.3 of *The General Divisor Theorem*. A weighted-vector construction and a diagonal point were previously given a shared label, but direct evaluation gives different angular readings. The example motivates the distinction

shared label $\nRightarrow$ shared reading $\nRightarrow$ shared specification.

The purpose is methodological: a numerical reading should be stated together with the specification that produces it, and equality of readings alone does not specify a structural correspondence between the underlying objects.

1. Weighted-Vector Example

An earlier draft defined a weighted tuple $(9, 4, 2, 3)$ with angles $\theta_k \in \{0^\circ, 60^\circ, 120^\circ, 180^\circ\}$ and

$$V = \sum_{k=1}^4 w_k e^{i\theta_k} = 9 + 4e^{i\pi/3} + 2e^{i2\pi/3} + 3e^{i\pi} = 7 + 3\sqrt{3}i.$$

The reported reading was $r_V = \arg(V) \approx 36.586776^\circ$.

The same draft separately noted that the point $(1, 1)$ has reading $r_D = \arg(1 + i) = 45^\circ$, and described the combination as a “9423 phase-lock representation,” implying a correspondence between the two constructions.

Since $r_V \neq r_D$, no such correspondence holds under the stated construction.

An exhaustive search over all 24 permutations of assigning the four weights to the four fixed angles confirms that this discrepancy is not an artifact of ordering: no permutation reaches exactly 45° , with the closest approaches at 43.898° and 46.102° (`tests/basis/test_9423_phase_lock.py`).

This gives the three-level distinction

shared label $\nRightarrow$ shared reading $\nRightarrow$ shared specification.

The draft’s own label, “9423 phase-lock representation,” was shared across both objects by assertion, but the named readings are not shared: $r_V = 36.586776^\circ \neq 45^\circ = r_D$.

The second implication is worth stating independently. Even had $r_V = r_D$ exactly, that equality alone would not establish a structural correspondence between V ’s construction and the point $(1, 1)$. No transformation, projection, or limiting operation connecting the two objects would thereby be specified.

A shared label asserts a correspondence. A shared reading is, at most, evidence that two objects were run through procedures producing the same observable category. Neither, on its own, specifies the correspondence itself.

Section 3 gives a cleaner instance of the same principle. The reported ratio for $m = 6, a = 5$ has taken three different numerical values during the paper’s development: $24/25$ (originally reported, against no explicitly stated refined predicted count), 3 (against the naive predicted count $(\varphi(P_k)/P_k)(L/6)$), and 1 (against the refined predicted count $L\varphi(P_k/6)/P_k$). None of these numbers is meaningful without specifying which predictor produced it. The number alone is insufficient; the reading requires its producing specification to be stated alongside it.

2. Mathematical-Basis Reading

The example separates three claims that can otherwise be conflated. A shared label is an asserted identification. A shared reading is equality of an observable produced by stated procedures. A shared specification requires an explicit mathematical connection between the underlying constructions, such as a transformation, projection, equivalence, or other defined map.

Accordingly, a reading is most informative where its producing specification is stated alongside it. Different predictors or constructions can attach different meanings to the same reported number, while equal numbers can arise without establishing a correspondence between the objects that produced them.

3. Verification

The exhaustive permutation check referenced above is implemented in `tests/basis/test_9423_phase_lock.py`. The check evaluates all 24 assignments of the weights $(9, 4, 2, 3)$ to the four fixed angles and confirms that none produces exactly 45° .

Conclusion

The worked example supports a compact specification rule:

state the reading together with the specification that produces it.

A shared label can motivate a comparison. A shared reading can provide evidence for further comparison. A shared specification requires the mathematical connection itself to be stated.